

Road Traffic Mortality Rate Analysis Saudi Arabia

*Tracking progress on SDG Indicator 3.6.1
and Vision 2030*

Prepared by: Zainab, Data Analyst (2025)

Portfolio: <https://my-portfolioza.netlify.app>

1. Project Overview

What we are measuring: SDG Target 3.6.1 aims to cut global road traffic deaths and injuries in half by 2030. This report tracks Saudi Arabia's progress since 2016, compares it to WHO benchmarks, and calculates how close the Kingdom is to the Vision 2030 target of under 10 deaths per 100,000 people.

Where the data comes from: The original file only had two numbers: an international estimate of 18.5 (UN Stats, 2021) and a national rate of 12.13 (Health Sector Transformation Report, 2024). I used the source links embedded in that file to dig deeper. I rebuilt the 2016–2024 trend using the HST 2024 chart, confirmed the 2016 baseline of 28.8 via WHO news, and got GCC/Arab country comparisons from the 2023 WHO Global Status Report. Local regional data comes from a 2025 study in Scientific Reports / Nature that used the Saudi Open Data Portal. Every data point on the dashboard is labeled so you can easily tell verified official figures apart from trend estimates.

2. Key Findings & KPIs

A 57.88% Drop in Deaths

Saudi Arabia's road traffic death rate fell from 28.8 in 2016 down to 12.13 in 2024. This is one of the most impressive drops in the world for an eight-year period, especially for a country with such a massive highway network.

Below the Global Average

The current national rate (12.13) is now 19.13% lower than the WHO global average of 15.0. Moving below the global average shows that the Vision 2030 road safety programs are bringing permanent, structural improvements.

4,608 Lives Saved Every Year

Total annual road deaths dropped from 9,031 in 2016 to 4,423 in 2023. That is the real human impact behind the statistics—4,608 fewer deaths a year.

2.13 Points to the Finish Line

The Vision 2030 national target is a rate below 10.0. Since the current rate is 12.13, we only have a 2.13-point gap left to close. Saudi Arabia has already finished about 88.6% of the journey from the 2016 baseline to the 2030 target.

3. Regional & International Context

GCC and Arab Country Comparison (2021 Data)

In the Gulf region, the UAE (5.9) and Qatar (7.3) have the lowest mortality rates, followed by Bahrain (8.1) and Kuwait (9.2). Saudi Arabia's 2024 rate of 12.13 puts it ahead of Oman (11.0) and significantly ahead of Jordan (13.6). When you factor in Saudi Arabia's massive geographic size, long desert highways, mountain routes, and heavy logistics traffic, this is a remarkably strong result.

Disparities Between Saudi Regions (2023 Data)

The national average doesn't tell the whole story. Tabuk (23.4) and Jazan (20.4) have the highest death rates in the Kingdom—more than double the rates of Riyadh (11.1) and the Eastern Province (10.3). Interestingly, Al Baha (147.4) and Madinah (145.4) have the highest injury rates per 100,000 despite average death rates. This means many high-speed crashes there are severe but survivable, which points to a different type of risk that needs a targeted medical response.

4. Why are Some Regions Higher Risk?

Long Desert Roads and Tired Drivers

Tabuk connects distant towns via long, single-lane desert highways. Driving long hours at high speeds across empty terrain naturally increases the chances and the severity of crashes.

Tricky Mountain Roads

Jazan and Al Baha feature steep slopes and sharp curves through mountain areas. These roads leave zero room for mistakes and get very dangerous during heavy fog or bad weather.

Fewer Speed Cameras

The Saher automated system is what drove the national reduction. Big urban areas with plenty of cameras—like Riyadh (11.1) and the Eastern Province (10.3)—have the lowest death rates. On the flip side, remote regions with fewer cameras record the highest rates.

Slower Emergency Response Reach

In deep mountain and desert areas, it takes much longer for an ambulance to reach a crash scene than it does in a city. This delay directly affects survival rates, explaining why some regions with a moderate number of crashes still have a high number of deaths.

5. Practical Recommendations

Expand Speed Cameras to Remote Routes

Moving Saher cameras to the underserved highway stretches in Tabuk and Jazan is the most reliable way to lower deaths. Adding both fixed and mobile cameras on inter-city desert routes will directly target the main issue: speeding on open roads.

Upgrade Safety Engineering on Mountain Roads

Invest in stronger guardrails, high-visibility lane markings, solar-powered lighting, and reflectors in the mountains of Jazan, Al Baha, and the Tabuk highlands. This will make these dangerous routes much safer.

Add More Air Ambulance Landing Spots

Set up extra helicopter landing zones near the most frequent crash sites in Jazan, Al Baha, and Tabuk. This cuts down the time it takes to get severe crash victims to a trauma center where ground ambulances face slow routes.

Set Up a National Road Safety Observatory

Create standard, annual regional reporting for both death and injury rates. This will help us track SDG 3.6.1 progress and show exactly where resources need to go each year.

Share the Saudi Success Story Internationally

Dropping traffic deaths by nearly 58% across a massive country is a major global achievement. Documenting and publishing how Saudi Arabia combined automated enforcement with good governance can serve as an excellent blueprint for other nations with large highway networks.